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Optimizing ESP Performance in High GVF Wells: A Case Study

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Abstract

This paper addresses the challenges of artificial lift in gassy formations by revising Electric Submersible Pump (ESP) designs to accommodate unexpected reservoir behavior. Initially targeting a static pressure of 8500 kPa and a productivity index (PI) of 0.05 m³/d/kPa, post-drilling evaluations revealed lower pressure and resulted higher gas volume fraction (GVF). To counter these setbacks, ESPs were repositioned below the perforation interval and reinforced with preventive hardware, such as cable guards and blast joints, to enhance system integrity.

Analysis identified high GVF intake as per primary case performance. To mitigate this, the ESP depth was adjusted from to below perforation to reduce gas ingestion, enhance hydrostatic head, and improve motor cooling. Additional preventive measures, including metallurgy compatibility checks and a packer installation 500 m above the ESP, were implemented to manage gas accumulation. These modifications reduced GVF, improved static intake pressure to 8500 kPa, and increased net oil production by over 300%. The ESP run life extended by 200%, significantly reducing downtime costs by \$13,200 per day.

Introduction

New formation within the operating field, first targeted in 2024, faced significant artificial lift challenges due to unexpected reservoir behavior. Initial designs assumed higher static pressure and a productivity index, which resulted in a suitable gas volume fraction entering the pump and expectation of relatively stable flow. Thus, ESPs with Multiphase Gas Handler (MGH) and Advanced Gas Handler (AGH) were chosen. However, post-drilling, static pressures found less than expected and productivity index dropped less than expected resulting in higher separation of gas and leading to unstable flow that exceed the capacity of used gas handling string. Four wells commissioned under these conditions experienced frequent ESP trips, high motor temperature shutdowns, and unstable amps, drastically reducing run life and production. New drilling wells could deviated from initial expectations in most of the time and the capability of the completion string will work to fit the requirement due to the flexibility of the used resources in the well mostly the variable speed drive, in our case Esp reached to the maximum capacity of the gas handling which cased frequent trips in the commissioned wells that leads to nonproductive time.

In ESP operation when a certain amount of free gas is presented at the pump inlet it starts to take the usable volume within the pump which restricts the volumetric efficiency of the pump (Karansov 2014). That results in a reduction in the expected production and the generated head from the pump stages. In longer periods that cumulative gas bubbles may fill the stages completely and stop the pump from lifting fluid and cause no flow at some point. Such challenges of degradation due to high GVF are discussed deeply in literature like (Hua, 2012) showing the effect This condition is defined as a gas lock. (Fig.1)

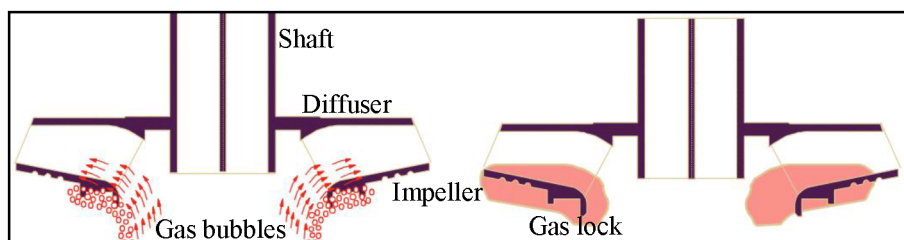


Figure 1—gas interference and gas lock

The design of pump stage types has a remarkable effect on the ability to handle the amount of gas in the pump. Radial stages could handle 10% to 15% free gas volume fraction and the mixed flow stages can handle 15% to 25 % free gas volume. Modification of the impellers were proposed in some applications where high GVF confirmed as solution as discussed (Badr, 2021). Depending on the expected gas volume selection of mixed flow in gassy application took place in our case as ESP were completed with (D1050N) 400 series pumps with two sections of 121 mixed flow stages with RLOY hosing and INCONEL shaft.

The amount of gas could be either separated by the gas separators where the fluid separated by the centrifugal effect inside the gas separator tool ending with gas directed in separate bath out from the fluid or enhanced natural separation methods where gas separated out from fluid by forcing the gas to go through a bath that increases the natural separation due to the relation between pressure and depth. Both methods targeting gas displaced back to the annulus to be produced from the casing, or to be handled by different handling devices that have been proven in the Esp market, which is called gas handlers. It works to compress the amount of the free gas back into fluid, homogenize the mixture by reducing the bubble size and simplify gas movement into the mainstream flow

challenges

In the initial stage during the planning of the ESP completion for the new formation wells ESP was supposed to face an acceptable amount of fluid with number of gases that could be handled using handling devices. The selection was depending on resulted gas volume fraction that will enter the pump by the expected draw down as per production requirement and static pressure expectations, so in the initial case of wells with pump setting depth above perforation the parameters were suspected to result in 34% GVF entering the intake and to produce the required gross by 50000 kpa draw down. Fig.2 demonstrated the expected base case of the running ESP during the design stage.

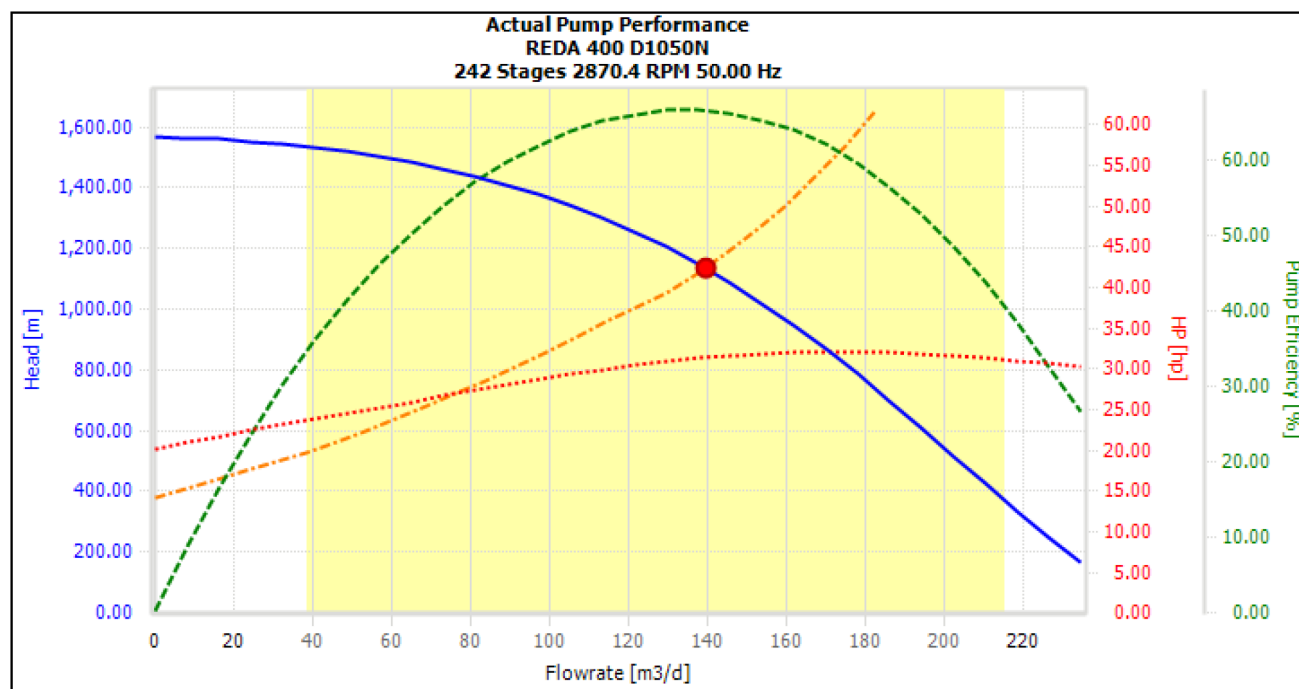


Figure 2—pump curve during design stage

Considering that those wells were designed with expectation of gassy formation the selected pump design were mix flow stages to handle the combination of gas and fluid through the interior bath of the selected pump, D1050N pump shows good performance in the gassy fields with close gross requirement within the region. That enhanced the selection along with considering advanced gas handler that is capable to compress the gas back into fluid.

Once the installation takes place the initial recorded static pressure by the installed gauge shows lower static pressure than the expected pressure during the planning stage, that resulted in more gas separation as we are operating below bubble point, considering that GOR also was not confirmed yet as this formation was targeted first time targeted without a historical separator test from same formation.

While commissioning the well it turned out that to get flow at surface considering same frequency on planning stage will lead to higher draw down in well dynamic pressure while operating the ESP, more gas separated out of fluid resulted in higher GVF entering the pump causing gas locks and no flow conditions. ESP operated with lower offtake to decrease the effect of gas separation due to low dynamic pressure.

Extra resources were assigned on trouble shooting the well taking effort and time of the ESP field team to stabilize the wells that affected the other activity in the same cluster were clearly raising after commissioning the first 4 wells. Several meetings were specified for analyzing and discussion of those wells with deferent parties from ESP provider and client side.

Troubleshooting

Operating conditions. Troubleshooting efforts were undertaken, and various strategies were tested to address the operational challenges. Initially, wells were operated at high Tubing Head Pressure (THP) to compress the gas back into the fluid, as fluctuating running amps were observed due to gas locks and unstable inflow behavior. Subsequently, wells were operated at lower frequencies to reduce offtake, aiming to maintain higher dynamic intake pressure, which was thought to potentially stabilize operating parameters.

The Variable Speed Drive (VSD) capabilities were utilized to enable a smart operating mode that could adapt to variations in inflow. Both pressure regulation mode and current regulation mode were configured and fine-tuned to enhance flexibility in response to inflow changes. Despite these adjustments, the trials

were unsuccessful because the inflow was unable to be managed due to the high gas volume fraction, well keep tripping on high motor temperature due to no flow and gas lock. The frequency often reached the maximum allowable limit of the settings to sustain the targeted pressure or targeted current without successful restoration of flow, in most of the trails flow could not recover due to the excessive presence of gas inside the pump and intake particularly reflected during periods of low load and decline of THP.

Attempts were also made to operate at higher frequencies to expel gas from the pump, with the expectation that fresh fluid from the formation would enter the pump, thereby reducing the gas volume fraction. However, this approach resulted in a significant drop in intake pressure, which caused more gas to evolve out of the solution. This, in turn, led to conditions such as motor overheating or underload scenarios.

Wireline. External resources were sought to improve the troubleshooting process. A Sliding Sleeve Door (SSD) was utilized to release gas that had accumulated below the packer, even though the packer was set 500 meters above the Electric Submersible Pump (ESP). This setup was intended to allow the long wellbore section below the packer to act as a reservoir, delaying the breakthrough of accumulated gas into the intake. Additionally, backflushing was applied, drawing on its successful use in other clusters where it homogenized the downhole fluid flow and mitigated gas lock issues.

Despite these efforts, both the SSD and backflushing trials yielded similar results, with the well showing no significant improvement in performance. This indicates that the root cause of the issue is likely related to the gas being released from the inflow originating from the formation itself

Injector wells. Enhancing the pressure of the producing well through the Effort of improving the injector wells surrounding the subjected well was attempted and discussed during the process of trouble shooting. Minor result was recorded by down hole gauge and that mostly due to the tight formation that limits the effect of the injection.

In conclusion well was running for few hours during each restart leading to a trip on either low cooling or low load, regardless of the applied improvements the static pressure reached to a stable level which still not enough to maintain the limitation of the ESP string after drawing down in the operating condition. Thies restarts led to only around 7 days up time in ESP trends as per Fig3 and in total most of the operating time was a no flow condition. (Fig.3)

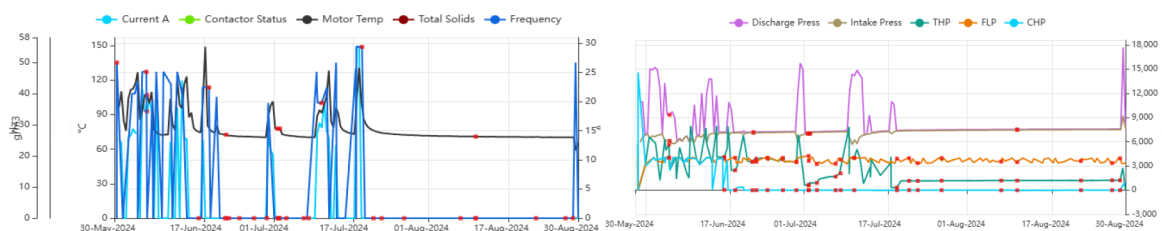


Figure 3—trends for well with PSD above perforation, well was running for 7 days in the first 3 months

Planning the new installation

After experiencing performance setbacks, a root cause analysis identified high GVF intake as the main issue. Engineering teams proposed two solutions: (1) removing the packer for casing-based gas separation or (2) installing the ESP deeper below the perforation to increase hydrostatic head and reduce GVF that will enter the intake. HSE limitations was the main drive for selecting option two due to the followed guidelines. Guidelines of the new formations planning procedure stated that if the well flowed for more than 72h without support from water injector, packer must be used to work as barrier against the flow as part of the HSE practice regulation, considering that this formation was expected to have good potential and there is chance of exhibit self-flow under high-case scenarios, so the drilling planned with completion packer to isolate the production zone. During the period of the commissioning and the startups of the first few wells there were no grants that there is no expectation of high-pressure expectation that could cause a

risk, so packer removal was not a choice considering that drilling of the planned wells in this formation was planned within second half of 2024. Packer removal could be implemented after reevaluation and modeling the running parameters depending on the classification of the well by converting the absolute open flow into kg/s and comparing it with classification. Approval from the integrity team after review also takes place to re classify the well according to the new modeled data that proved the new formation has lower initiative compared to the expectation.

Drilling of all wells in this scope was planned with a perforation wash and chance to evaluate if HCL stimulation is required. During the drilling of the well's acid stimulation conducted for wells with low water injection rate during rig operations using HCL to enhance injectivity ensures clean production environment.

The targeted formation was a tight, highly competent carbonate matrix with a limited degree of fracturing. The oil column seems to be underlain by water but, aquifer support appears to be slight and water ingress into the producing wells was limited. Reservoir rock properties are good with porosities are typically 15% to 25% and the vertical interval in the crestal area above the oil water contact is about 40m to 50m

Methodology

Modeling was run in with historical data from recent drilled wells shows that by lowering ESP below perforation the static bottom hole pressure will increase to a point that leads to less gas separation during the period of production leading to getting low gas into pump with the help of gas handler devices. Also, as part of protection for the string and to maintain good protection for the electrical integrity cable guards were planned as part of the design to handle the possibility of jetting effect across perforation zone. String selected to be RLOY housing, head and base, that fits the expected down hole condition in terms of pressure, temperature, partial pressure for the different fluids as per available PVT data analysis. Implementation in well A is done by lowering ESP completion below the perforation while using a shroud to force the flow down to enable motor cooling. This gives the chance for the free gas to go up in the annulus and a lower GVF fluid will enter the pump. Completion assembly was planned properly during the preparation to ensure that splice between the motor lead extension to the main cable is located below the perforation zone as cable guard will not fit in the outer diameter of the splice. Design went through approval review and modification through meeting with engineering department within the field operator company to ensure no gaps within design and it was the first implementation in this field. Wide range of uncertainties were implemented in three case scenario design to cover the expected gas oil ration from 40 m³/m³ to 200 m³/m³, static pressure from 6500 kpa to 8500 kpa, water cut from 40% to 90% and productivity index from 0.02 m³/d/kpa to 0.005 m³/d/kpa. Tally reviewed during the installation to confirm locating the cable guard across perforation and safety factor considered 10m above and below the perforation zone. Designing the string took into consideration that ESP was set with acceptable space from well bottom to avoid any solids or formation immigration through the fluid bath within the pumps and intake.

Results after implementation

The results after the implementation of the improvement in the first well showed remarkable improvement and initial signs of having stable production during the first monitoring days were clear, improvements affected:

Static pressure. Static intake pressure improved from 5600 kpa in the previous completed wells to 8500 kPa (in the beginning of Fig-4 trend) recorded by downhole gauge during the first attempt to restart during the commissioning of the well. While operating the ESP and production at surface dynamic intake pressure was 5800 kpa as recorded in Fig-8, compared much lower values in similar wells that completed with ESP setting depth above perforation and leaded to unstable running conditions. This gives the advantage of keeping the higher operating intake pressures after the draw down. In addition to that there was no stable

pressure in the previous wells all trials were unable to maintain continuous flow due to the gassy condition that ESP faced during operation.

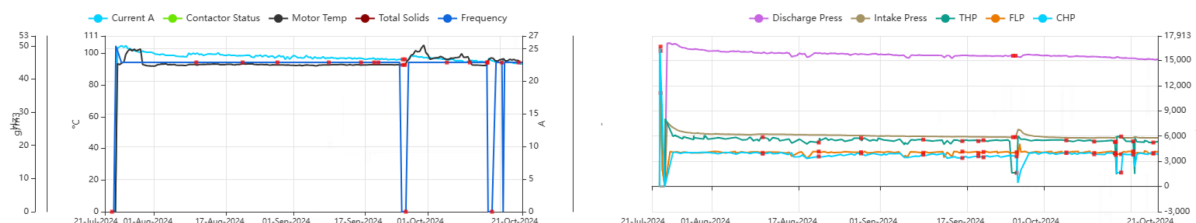


Figure 4—trends for well with PSD below perforation, well was only 5 days down due to flow line trips

Gas volume fraction. GVF shows a high amount of gas reaching 85% in the case of pump setting depth above perforation that calculated by the modeled data as well as was not flowing for long period. Which result after the gas handler 38.64% GVF in the lower stages of the lower pump (Fig-5), that percentage is higher than the handling capacity of ESP. In the actual data modeling for the case of ESP setting depth below perforation around 30% gas volume fraction was entering the intake before separation, which can be handled with the aid of gas handling and resulting in 5.76% GVF in the lower stages of lower pump (Fig-6). Data were simulated for the cases of ESP setting depth above perforation considering the data sheet expectation in term of fluid data as there was no test for the well due to unstable operation that limits the testing of non-running wells.

Outlet	
Pressure	3672.48 kPa
Liquid Rate	66.687 m3/d
Gas Rate	42.000 m3/d
Gas Volume Fraction	38.64 %

Figure 5—GVF at first stage of lower pump for ESP setting depth above perforation

Outlet	
Pressure	8133.74 kPa
Liquid Rate	175.851 m3/d
Liquid Rate	10.739 m3/d
Gas Volume Fraction	5.76 %

Figure 6—GVF at first stage of lower pump for ESP setting depth below perforation

Plotted GVF data against the pressure at intake depth for the previous compared case gives better visualization for the expected gas in different targeted draw down conditions. Random selected pressure (4500 kpa) reflects a significant change in the below Fig.7 and Fig.8

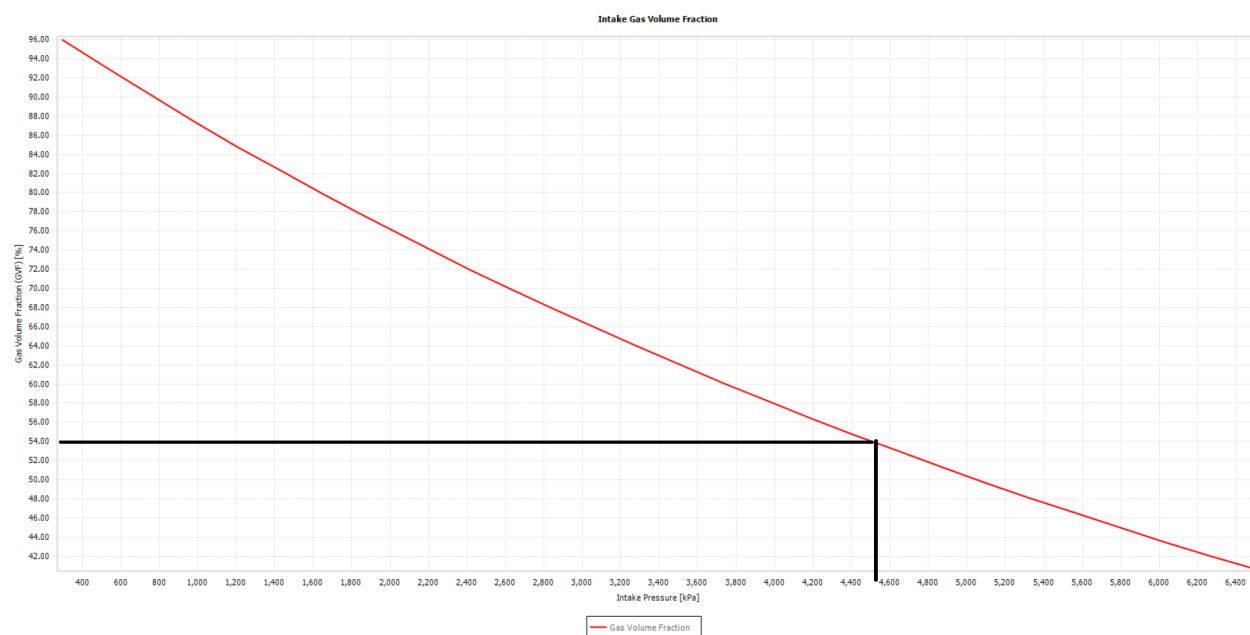


Figure 7—GVF VS intake pressure for ESP setting depth above perforation

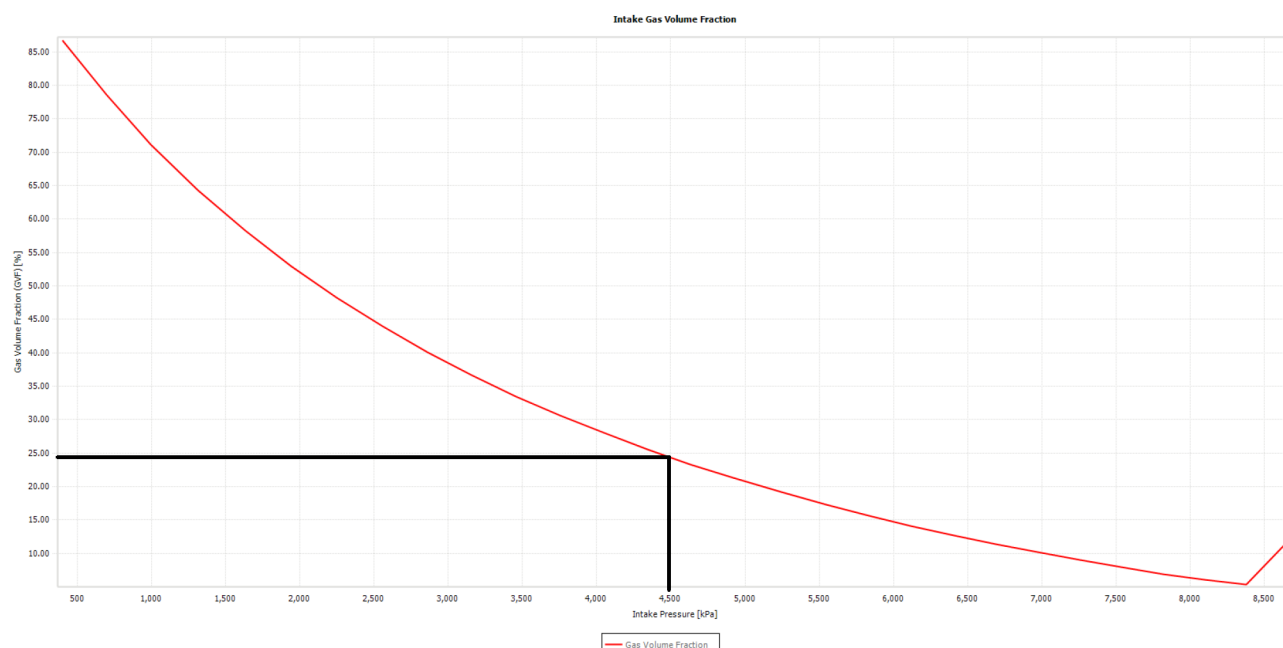


Figure 8—GVF VS intake pressure for ESP setting depth below perforation

We did a modeling to in the same well after validating the data in the new well with setting depth below perforation, the plan was to see the effect of the setting ESP deeper on the GVF as this reduction could be due to different factors, so we modeled two case in same well and showed that around 25% of GVS reduced due to setting the ESP deeper and the rest are resulted due to the other changes in pressure and productivity. Fig.9 and Fig.10 showing the effect of GVF reduction on the first stage of the pump after modeling the same fluid parameters of running well.

Outlet	
Pressure	5062.81 kPa
Liquid Rate	155.979 m3/d
Liquid Rate	73.144 m3/d
Gas Volume Fraction	31.92 %

Figure 9—GVF at first stage of lower pump for modeled ESP setting depth above perforation

Outlet	
Pressure	8133.74 kPa
Liquid Rate	175.851 m3/d
Liquid Rate	10.739 m3/d
Gas Volume Fraction	5.76 %

Figure 10—GVF at first stage of lower pump for ESP setting depth below perforation

Production. Wells were having a good expectation during the planning period, but real data showed that wells were doing nearby the low case scenario and operation of wells were unstable which resulted in intermittent periods of low flow to surface, implementation of the improvement resulted in a stable continues flow and net oil production increased to more than 300%. Which is the key factor that supports the implementation of setting ESP below packer.

Extra cost. Downtime cost savings reached \$13,200 per day avoided. Failed strings were planned for well intervention after unsuccessful trials ended with mechanical or electrical failures with an average of 90 days. Avoided NPT of average 100 days between the failure date to the commissioning of the new string. The sustained production improved the financial measures, and a significant reduction in the payback period reached 44 days considering the unit technical cost.

Reliability. As the previous wells suffered from poor flow conditions, wells faced failures with different scenarios either electrical or mechanical failures due to no flow conditions as later proofed during tear down. In addition to that wells were off most of the time for pressure build up between trails and off waiting for gas bleed off. Implementation ensured that ESP run life extended by 200% compared to similar wells in the field (ESP still running healthy)

Resources utilization. Defective resources due to the trouble shooting were clear after replacement of failed strings such as broken shafts, melted cable insulation due to overheat during the no flow conditions. In addition to that permanent magnet motor (PMM) were un effectively utilized in some of the wells which were supposed to result in good power saving for the long-term consideration. Work over hoist resources were directed to another scope withing the field that improved the overall performance. Man-hours were utilized

Well shows stable operation after the implementation and trends were stable as in Fig10, during operation flow up, we observed that gas accumulation shows some effect in the running parameters effect after 30 days and fluctuation in the motor amps along with THP fluctuation were started to appear although packer was set 500m above the ESP, so proactive SSD gas release scheduled every 30 days to prevent gas accumulation below the packer by wire line unit to make sure no trips could occur due to gas locks or poor colling that leads to downtime and production losses. Implementing this solution ensures no further instability observed post-implementation.

Conclusions

In our project, we focused extensively on ESP component selection by simulating the expected gas calculations to determine the optimal gas handling method. This careful analysis clearly demonstrates

the advantage of placing ESPs below the perforation in high GVF wells with mandatory packer system requirements. By setting the ESP deeper, we increased recovered production per drawdown and reduced gas ingestion, which preserved motor function and stabilized amp draw. The use of preventive hardware, such as blast joints and cable guards, proved essential protection for maintaining long-term integrity.

From an economic standpoint, the project resulted in a remarkable win due to daily production gains and a payback period reached up to 44 days after sustained production along with the avoidance of costly interventions. Moreover, the strategy effectively mitigated the logistics involved in ESP replacement, avoiding downtime that previously extended beyond 100 days until production was restored with a new installed ESP.

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